Sliding Window Or

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the bitwise or of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k: the number of elements and the size of the window.

The next line contains four integers x, a, b and c: the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window ors.

Constraints

- $1 \le k \le n \le 10^7$
- $0 \le x, a, b \le 10^9$
- $1 \le c \le 10^9$

Example

Input	Output
8 5 3 7 1 11	4

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their ors are 11, 15, 15 and 15. Thus, the answer is $11 \oplus 15 \oplus 15 \oplus 15 = 4$.